

ENGINE KNOWLEDGE DOCUMENT

Integrated Drive and Powertrain Engine System

1. Introduction

This document describes an integrated drive and powertrain engine system used in heavy-duty industrial and construction machinery.

2. Purpose

The engine provides high torque, hydraulic power, efficient transmission, and safe braking.

3. Major Components

Pump Motor: Drives the hydraulic pump.

Hydraulic Power Module: Supplies pressurized fluid.

Traction Module: Enables vehicle movement.

Drive Motor: Main rotational power source.

Planetary Gear System: Torque multiplication and speed reduction.

Wet Disk Brakes: Maintenance-free braking system.

Cast Steel Housing: Protects internal components.

4. Working Principle

Electrical power drives the motor, gears adjust torque and speed, hydraulics supply pressure, brakes ensure safety.

5. Applications

Construction machinery, mining vehicles, industrial transport systems, AGVs.

6. Keywords

planetary gear system, wet disk brakes, hydraulic module, traction motor